
Validity-Gated Counterfactual Consistency Regularization for Korean Hate Language Detection

Korea University COSE461 Final Project

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Abstract

Hate and offensive language classifiers often treat identity terms as shortcuts, but counterfactual mitigation methods usually assume that replacing one identity mention with another preserves the label. This assumption is unreliable in Korean, where identity substitutions interact with morphology, post-positional particles, slang, and context-specific social meanings. We propose validity-gated counterfactual consistency regularization, a Korean-specific framework that separates identity counterfactual generation from validity assessment before using counterfactual pairs as training signals. We evaluate the method on K-HATERS with KLUE-RoBERTa, comparing standard fine-tuning, ungated identity-swap CCR, strict validity-gated CCR, a coverage-matched strict variant, a masking sanity baseline, and fixed-lambda strict ablations. In the controlled main comparison, the coverage-matched strict variant improves StrictPairAcc from 0.8076 to 0.8295 over the baseline and from 0.8171 to 0.8295 over ungated Naive Swap, while preserving Macro-F1. A stronger strict setting further reaches 0.8381 StrictPairAcc. These findings suggest that, for Korean hate and offensive language detection, counterfactual regularization should account for whether identity substitutions are actually label-preserving.

1 Introduction

Hate and offensive language detection is often formulated as a standard text classification problem, but high aggregate accuracy can hide systematic failures around identity terms. A Korean hate speech detector may learn that words referring to gender, ethnicity, religion, age, or sexuality are strong shortcuts for offensiveness, rather than relying on the actual abusive intent of the sentence. As a result, a model can flip its prediction when only the identity term is replaced, or falsely classify neutral identity mentions as hateful.

Counterfactual consistency offers a natural way to test and reduce this behavior. If two sentences differ only in the identity term and should preserve the same label, the model should make consistent predictions for both. However, applying identity swaps naively is especially risky in Korean. Simple lexical substitution can break morphology, create awkward slang forms, or leave identity-specific context unchanged. For example, replacing “폐미” with “한남” can produce unnatural forms, while replacing “이슬람” with “기독교” may leave context such as “이슬람화” semantically inconsistent. Such pairs are not harmless noise: using them for training can teach the model invalid invariances.

In this work, we study *Validity-Gated Counterfactual Consistency Regularization (CCR)* for Korean hate and offensive language detection. We fine-tune KLUE-RoBERTa on K-HATERS and compare a standard classifier, ungated identity-swap consistency regularization, and validity-gated variants. Our

method first generates Korean identity counterfactual pairs, then filters them with Korean-specific validity conditions before applying a symmetric KL consistency loss between the original and counterfactual predictions.

Our experiments show that counterfactual consistency improves robustness while preserving base task performance. In the main comparison, Strict-Matched achieves 0.7906 Macro-F1 and 0.8295 StrictPairAcc, compared with 0.7882 and 0.8076 for the baseline and 0.7916 and 0.8171 for ungated Naive Swap. These results indicate that validity filtering can improve correctness-aware counterfactual robustness without substantially reducing standard classification performance, supporting our hypothesis that more counterfactual pairs are not always better when some provide invalid training signals.

Our contributions are threefold. First, we propose a Korean-specific validity-gated counterfactual regularization pipeline for hate and offensive language detection. Second, we compare ungated and gated counterfactual training under the same model and dataset setting. Third, we analyze the trade-off between counterfactual coverage and validity, showing that stricter filtering can improve correctness-aware robustness without sacrificing standard classification performance.

2 Related Work

Korean hate and offensive language detection. Several Korean hate speech datasets have been proposed to support hate and offensive language detection. K-HATERS provides a large-scale Korean dataset with fine-grained labels for hate and offensive expressions, making it suitable for supervised hate/offensive classification [1]. KOLD focuses on Korean offensive language in social media and highlights the difficulty of annotating implicit and context-dependent offensiveness [2]. K-MHaS further studies multi-label Korean hate speech across multiple target categories [3]. Our work uses K-HATERS as the main training and evaluation dataset, but focuses on a different question: whether models trained on Korean hate data remain consistent under identity-term counterfactuals.

Identity bias in toxicity classifiers. Prior work has shown that toxicity and hate speech classifiers often rely on identity mentions as spurious cues. Dixon et al. demonstrate that toxicity models can assign high toxicity scores to neutral sentences containing identity terms, revealing unintended bias in text classification [4]. This problem is closely related to our setting: Korean identity terms can also become shortcuts for hate prediction, especially when some groups appear more frequently in offensive contexts. Therefore, evaluating only standard F1 is insufficient; models should also be tested for counterfactual robustness under controlled identity substitutions.

Counterfactual fairness and data augmentation. Counterfactual data augmentation has been used to reduce spurious correlations by modifying sensitive attributes while preserving the task label. Garg et al. propose counterfactual fairness for text classification through identity substitutions, encouraging models to behave consistently across demographic variants [5]. Wang and Culotta use counterfactual examples to reduce spurious correlations in NLP models [6]. More recent work also studies identity-based augmentation for fairer text classification [7]. Our work follows this line of research but argues that Korean counterfactuals require explicit validity filtering. Unlike approaches that treat all identity swaps as valid, we separate counterfactual generation from validity-aware filtering and study how this affects consistency regularization.

3 Approach

3.1 Counterfactual Pair Generation

Counterfactual data augmentation reduces model reliance on spurious correlations [8]. We generate candidate counterfactual pairs through lexicon-driven identity-term swapping. A category-level lexicon specifies replacement pairs for identity categories such as gender, religion, ethnicity, age, sexuality, and disability; representative swap pairs are provided in Appendix A.3 and the full lexicon is available in the released code <https://github.com/angellashin/261RCOSE46101>.

Identity terms are detected using Kiwi tokenization [9] so that replacements respect Korean token boundaries. We perform token-aware substitution and morphology-aware post-positional particle

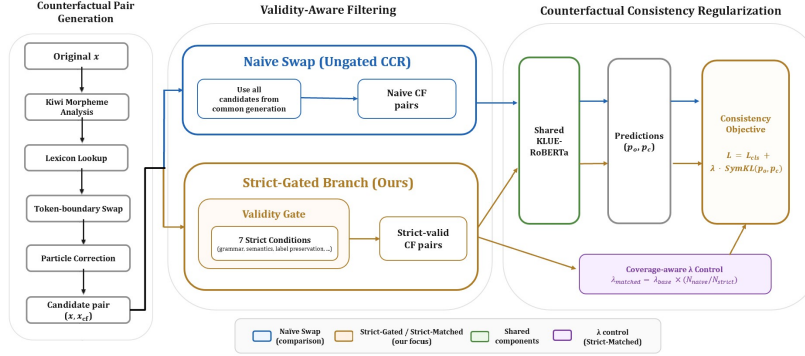


Figure 1: Overview of the validity-aware counterfactual consistency framework. **Naive Swap** uses all candidates directly; **Strict-Gated** applies a validity gate with 7 strict conditions before consistency regularization. **Strict-Matched** additionally scales λ to match the effective regularization strength of Naive Swap (Section 3.4).

correction to reduce grammatical artifacts, producing a candidate pair (x, x_{cf}) . This generation step is shared by all counterfactual training conditions: **Naive Swap** uses generated candidates directly, whereas **Strict-Gated** first applies the validity filtering procedure described in Section 3.2.

3.2 Validity-Aware Filtering

Candidate counterfactuals are not always label-preserving: an identity swap may change the factual, social, or pragmatic meaning of a sentence. We therefore apply validity-aware filtering before using counterfactuals for consistency regularization.

The strict gate filters candidate pairs using grammatical checks, semantic context filters, asymmetric-pair exclusions, comparison-token filters, context-specific filters for age- or incident-related expressions, and a harmful-object filter, accepting 5,964 of the 7,735 candidates. **Strict-Gated** applies consistency regularization only to counterfactuals that pass the strict gate; **Naive Swap** applies it to all lexicon-detected candidates.

3.3 Counterfactual Consistency Regularization

Following consistency regularization [10], we encourage distributional consistency between original and counterfactual predictions via a symmetric Kullback–Leibler penalty. For original predictive distribution p_o and counterfactual distribution p_c (softmax over the two classes),

$$\text{Cons}(p_o, p_c) = \frac{1}{2} (\text{KL}(p_o \parallel p_c) + \text{KL}(p_c \parallel p_o)).$$

The per-example training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \cdot \text{Cons}(p_o, p_c),$$

where $\mathcal{L}_{\text{cls}}$ is cross-entropy classification loss and λ is the nominal regularization weight. The consistency term is computed only for counterfactual pairs selected under the corresponding training condition.

3.4 Coverage-Aware Regularization

Because the consistency term is applied only to examples with valid counterfactuals, its aggregate influence depends on both λ and the counterfactual coverage $c = \#\{\text{valid CF training examples}\} / N_{\text{train}}$. We report the coverage-weighted regularization strength $\lambda_{\text{eff}} = \lambda \times c$, which summarizes the average per-example regularization pressure during training.

To separate coverage effects from gate-quality effects, we introduce **Strict-Matched**. Strict-Matched retains the strict validity gate but increases the nominal regularization weight so that

$$\lambda_{\text{matched}} \times c_{\text{strict}} \approx \lambda_{\text{base}} \times c_{\text{naive}},$$

where λ_{base} is the nominal weight shared by Naive Swap and Strict-Gated. This allows us to compare Naive Swap and Strict-Matched under similar aggregate regularization strength, so remaining differences more directly reflect which counterfactual pairs are selected by the gate.

Metric	Naive Swap	Strict-Gated	Strict-Matched
CF candidates	7,735	7,735	7,735
Pairs used for CCR	7,735	5,964	5,964
Train valid CF ratio c	0.04493	0.03464	0.03464
Configured λ	0.10	0.10	0.1297
Effective λ_{eff}	0.00449	0.00346	0.00449

Table 1: Counterfactual coverage and effective regularization strength. Strict-Matched matches the λ_{eff} of Naive Swap by scaling λ to compensate for lower coverage.

4 Experiments

4.1 Data

We use K-HATERS [1] (humane-lab/K-HATERS) with the repository splits: 172,157 training, 10,000 validation, and 10,000 test examples. Each input is a Korean text example, and the output is a binary abusive/offensive label. We map `offensive`, `l1_hate`, and `l2_hate` to the positive class and all other labels to the negative class. Inputs are tokenized and truncated to 128 tokens.

Counterfactuals are generated by lexicon-driven identity-term swapping as described in Section 3.1; representative pairs are provided in Appendix A.3. Validity filtering is applied before counterfactuals are used for consistency regularization. On the training split, the repository records 7,735 swappable candidates, of which 5,964 pass the strict validity gate.

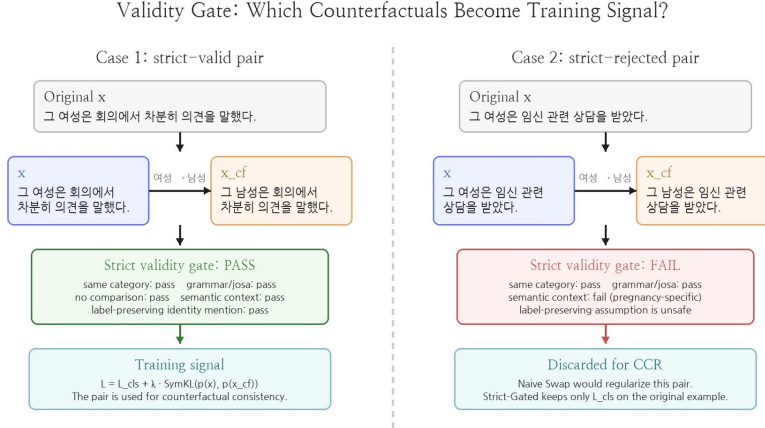


Figure 2: Strict validity gate applied to Korean gender-swap counterfactual pairs. Case 1 passes all criteria and contributes a symmetric KL consistency loss; Case 2 is rejected because 임신 (pregnancy) renders the label-preserving assumption unsafe for a gender swap, so only $\mathcal{L}_{\text{cls}}$ is applied.

4.2 Evaluation Method

Macro-F1 is computed as the unweighted mean of per-class F1 scores on the test set and serves as the primary measure of overall classification performance.

For pairwise counterfactual metrics, robustness is evaluated over the set S of swappable test pairs. For strict analyses, S is restricted to the strict-valid subset $S_{\text{strict}} \subseteq S$, consisting of pairs that pass all 7 strict validity conditions (Section 3.2). For an original example x and its counterfactual x_{cf} , the model produces a predicted label $\hat{y}(\cdot)$ and a positive-class probability $p(\cdot)$.

FlipRate. The flip rate is defined as

$$\text{FlipRate} = \frac{1}{|S|} \sum_{(x, x_{\text{cf}}) \in S} \mathbf{1}\{\hat{y}(x) \neq \hat{y}(x_{\text{cf}})\},$$

which measures how often the prediction changes under identity swapping. Lower values indicate greater robustness.

Probability Gap (ProbGap). Probability Gap measures the average change in model confidence between an example and its counterfactual. Lower values indicate more stable predictions.

Pair Accuracy. Pair accuracy is defined as

$$\text{PairAcc} = \frac{1}{|S|} \sum_{(x, x_{\text{cf}}) \in S} \mathbf{1}\{\hat{y}(x) = y \wedge \hat{y}(x_{\text{cf}}) = y\},$$

the fraction of pairs for which both the original and counterfactual examples are classified correctly. This metric jointly evaluates predictive performance and counterfactual consistency. The strict variants of all three metrics are computed identically over S_{strict} . For analysis, we further decompose $\text{PairAcc} = \text{Origcorrect} \times \text{consistency}$, where Origcorrect is the fraction of original examples classified correctly and *consistency* is the fraction of those correct originals whose counterfactual is also correct (Section 5.1).

4.3 Experimental Details

All experiments use `klue/roberta-base` with a single linear classifier on the CLS token. Models are trained with AdamW using a learning rate of 3×10^{-5} , batch size 64, maximum sequence length 128, weight decay 0.01, and 3 training epochs.

We run experiments with three random seeds: 42, 123, and 456. Reported results are averaged across seeds, and the best checkpoint for each seed is selected by validation Macro-F1. For the main comparison, we use $\lambda = 0.1$ for Naive Swap and Strict-Gated, while Baseline uses $\lambda = 0$. For Strict-Matched, λ is adjusted according to Section 3.4 to match the effective regularization strength of Naive Swap. We additionally run λ -ablation experiments for Strict-Gated with $\lambda \in \{0.05, 0.10, 0.1297, 0.15, 0.25\}$. For reference, the same ablation for Naive Swap is provided in Appendix A.5. As an additional comparison, **Masking Cons Reg** replaces the detected identity term with [MASK] and applies the same consistency objective without identity substitution.

4.4 Results

4.4.1 Main Comparison

Table 1 summarizes counterfactual coverage and effective regularization strength for each condition; Table 2 reports all metrics averaged over three random seeds.

All CCR conditions outperform the Baseline on correctness-aware robustness. Strict-Matched achieves the highest PairAcc (0.8264 ± 0.0031) and StrictPairAcc (0.8295 ± 0.0110), surpassing Naive Swap by 0.0096 and 0.0124 respectively, suggesting that validity filtering improves correctness-aware robustness beyond simply applying CCR to more pairs. Strict-Matched achieves this with 23% fewer CCR training pairs than Naive Swap, suggesting that cleaner pair selection can compensate for lower counterfactual coverage in this setting. Macro-F1 is stable across all conditions (0.7882–0.7917), confirming that CCR does not sacrifice classification accuracy for robustness.

Masking Cons Reg provides only a marginal PairAcc gain (+0.0081 over Baseline) and leaves FlipRate near the Baseline level (0.0491 vs. 0.0549). In contrast, all three swap-based CCR conditions reduce FlipRate to at most 0.024, confirming that the consistency penalty alone is insufficient without semantically contrastive pair construction.

Figure 3 visualizes these comparisons with error bars representing one standard deviation. Strict-variant metrics (S-FlipRate, S-ProbGap, StrictPairAcc) computed over the 350 strict-valid test pairs are reported in Table 6 in the Appendix.

4.4.2 Lambda Sensitivity

Table 3 reports Strict-Gated performance across $\lambda \in \{0.05, 0.10, 0.1297, 0.15, 0.25\}$, where $\lambda = 0.1297$ corresponds to Strict-Matched. ProbGap decreases monotonically from 0.0231 ($\lambda = 0.05$) to 0.0163 ($\lambda = 0.25$), while StrictPairAcc is non-monotonic, dipping at $\lambda = 0.10$ (0.8248) before peaking at $\lambda = 0.25$ (0.8381). Macro-F1 remains close across the sweep, suggesting that stronger strict regularization can reduce probability instability, although the best robustness setting may involve a small classification trade-off. Figure 4 illustrates the robustness–calibration trade-off.

Table 2: Main comparison results (mean $\pm$ std over 3 seeds). FlipRate and ProbGap: lower is better. Macro-F1, PairAcc, and StrictPairAcc: higher is better. PairAcc is computed over 455 robustness test pairs; StrictPairAcc is computed over 350 strict-valid test pairs. **Bold:** best per column; underline: second best. Strict-variant metrics are reported in Table 6.

Method	Macro-F1	FlipRate	ProbGap	PairAcc	StrictPairAcc
Baseline	0.7882 $\pm$.0026	0.0549 $\pm$.0112	0.0440 $\pm$.0049	0.8029 $\pm$.0102	0.8076 $\pm$.0128
Naive Swap	0.7916 $\pm$.0017	<u>0.0212</u> $\pm$.0041	0.0168 $\pm$.0006	0.8168 $\pm$.0037	0.8171 $\pm$.0023
Strict-Gated	0.7907 $\pm$.0035	0.0234 $\pm$.0090	0.0198 $\pm$.0012	<u>0.8220</u> $\pm$.0047	<u>0.8248</u> $\pm$.0088
Strict-Matched	0.7906 $\pm$.0041	0.0205 $\pm$.0037	<u>0.0176</u> $\pm$.0009	0.8264 $\pm$.0031	0.8295 $\pm$.0110
Masking Cons Reg	0.7901 $\pm$.0025	0.0491 $\pm$.0099	0.0435 $\pm$.0055	0.8110 $\pm$.0122	0.8133 $\pm$.0135

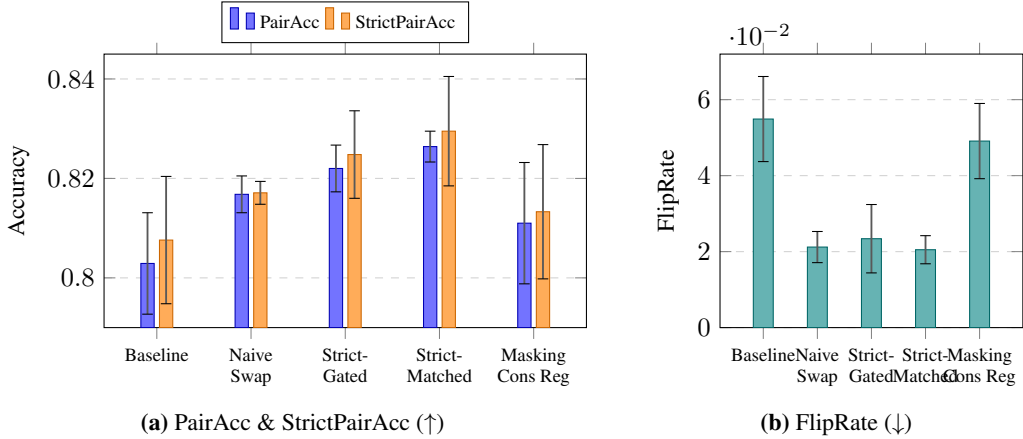


Figure 3: Robustness comparison across all five training conditions (error bars = 1 std over 3 seeds). Strict-Matched achieves the highest PairAcc and StrictPairAcc while maintaining a FlipRate comparable to Naive Swap, despite using 23% fewer training pairs.

5 Analysis

5.1 PairAcc as a Stricter Robustness Diagnostic Than FlipRate

Naive Swap achieves the largest absolute reduction in FlipRate relative to Baseline (-0.0337) yet trails Strict-Matched on PairAcc by 0.0096 . This divergence reveals a key limitation of FlipRate: it measures only whether the predicted label changes under swapping, without checking whether either prediction is correct. When invalid counterfactuals are included in training, the consistency pressure can force the model toward *consistent-but-wrong* predictions—a failure FlipRate rewards but PairAcc penalizes. We therefore treat PairAcc as the primary robustness diagnostic.

To understand what drives the PairAcc gap between Strict-Matched and Naive Swap, we apply the decomposition $\text{PairAcc} = \text{Origcorrect} \times \text{consistency}$ and attribute changes relative to Baseline. Both CCR variants improve PairAcc predominantly through consistency (Naive Swap: $+0.0332 \pm 0.0088$; Strict-Matched: $+0.0342 \pm 0.0084$), with a small offsetting decrease in per-example Origcorrect. The additional advantage of Strict-Matched over Naive Swap ($\Delta = 0.0095 \pm 0.0055$) traces almost entirely to the Origcorrect component ($+0.0086$, $\approx 90\%$), suggesting that the strict gate primarily removes pairs that suppress per-example Origcorrect rather than tightening pair-level consistency itself.

5.2 Lambda Sensitivity and Regularization Pressure

As shown in Figure 4, ProbGap decreases monotonically as λ increases, while StrictPairAcc is non-monotonic—dipping at $\lambda = 0.10$ before recovering to peak at $\lambda = 0.25$ (0.8381).

The $\lambda = 0.05$ sensitivity run also exceeds Naive Swap on StrictPairAcc, suggesting that the strict gate is not useful only because of coverage-matched scaling. Since this is an ablation run, we use it as supporting evidence rather than the headline comparison. This isolates the gate’s contribution from the coverage-matching step: even under lower aggregate regularization pressure, filtering label-unsafe pairs improves correctness-aware robustness. Strict-Matched ($\lambda=0.1297$) then adds

Table 3: Lambda sensitivity for the Strict-Gated family (mean $\pm$ std over 3 seeds). † Strict-Matched. ProbGap is lower-better; Macro-F1, PairAcc, StrictPairAcc are higher-better. **Bold:** best per column.

λ	Macro-F1	PairAcc	StrictPairAcc	ProbGap
0.05	0.7917 $\pm$.0042	0.8293 $\pm$.0099	0.8352 $\pm$.0144	0.0231 $\pm$.0031
0.10	0.7907 $\pm$.0035	0.8220 $\pm$.0047	0.8248 $\pm$.0088	0.0198 $\pm$.0012
0.1297 †	0.7906 $\pm$.0041	0.8264 $\pm$.0031	0.8295 $\pm$.0110	0.0176 $\pm$.0009
0.15	0.7917 $\pm$.0014	0.8249 $\pm$.0010	0.8305 $\pm$.0013	0.0168 $\pm$.0017
0.25	0.7898 $\pm$.0046	0.8300 $\pm$.0067	0.8381 $\pm$.0135	0.0163 $\pm$.0017

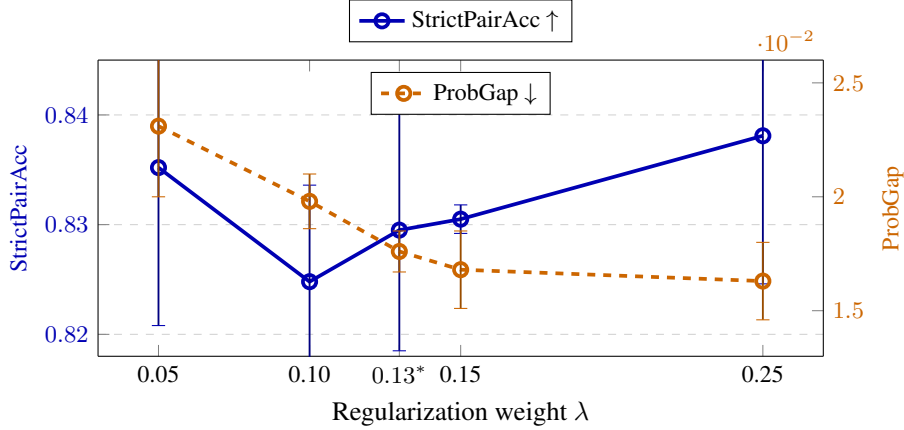


Figure 4: Effect of λ on StrictPairAcc (blue, left axis) and ProbGap (orange, right axis) for the Strict-Gated family. Error bars = 1 std. * $\lambda=0.13$ corresponds to Strict-Matched. ProbGap decreases monotonically while StrictPairAcc is non-monotonic, illustrating the robustness–calibration trade-off under stronger regularisation pressure.

a further coverage-normalized signal, confirming that gate quality and regularization strength are complementary rather than interchangeable.

To rule out the possibility that Strict-Gated’s advantage stems from asymmetric hyperparameter selection, we evaluate Naive Swap across the same $\lambda \in \{0.05, 0.10, 0.1297, 0.15, 0.25\}$ grid (Table 7). Strict-Gated consistently achieves higher StrictPairAcc at all five λ values ($\Delta_{\text{StrictPairAcc}} \in [+0.0009, +0.0123]$), demonstrating that the robustness advantage is stable across regularization strengths and cannot be attributed to differential tuning.

5.3 Contrastive Signal versus Regularization Signal

Masking Cons Reg applies the same nominal $\lambda = 0.1$ as the swap-based conditions but replaces identity tokens with [MASK] rather than semantically grounded substitutions. Despite identical regularization weight, it yields substantially smaller improvements in both FlipRate and PairAcc relative to Baseline compared with any swap-based condition (Table 2). This suggests that semantically grounded pair construction is an important condition in our setup for CCR to reduce identity-driven prediction instability, and that the robustness gains in swap-based conditions stem from the contrastive structure of the pairs rather than from the consistency penalty alone.

5.4 Qualitative Analysis

Table 4 presents five representative test pairs illustrating three behaviorally distinct cases. Gate-reject pairs (Examples 3–4) account for 105 of the 455 test pairs (23.1%); among the remaining 350 strict-valid pairs, the Baseline flips on 19 (5.4%), reduced to 7 (2.0%) after Strict-Matched training.

Gate-accept pairs (Examples 1–2). Both pass the strict validity gate yet induce prediction flips in the Baseline, confirming identity-term shortcut learning that CCR targets. Replacing 여성단체 with 남성단체 (Example 1) or 동성애자 with 이성애자 (Example 2) reverses the predicted label on otherwise identical context; all three swap-based CCR conditions eliminate these flips.

Table 4: Representative qualitative examples. Swapped terms are underlined. Examples 1–2: gate-accept pairs inducing Baseline flips. Examples 3–4: pairs rejected by the strict gate. Example 5: residual failure under Strict-Matched. All from seed 42.

#	Cat.	Original	Counterfactual	Y	Case
1	Gender	여가부.여성단체. 왁왁왁 #@이름# 미워할거양	여가부.남성단체. 왁왁왁 #@이름# 미워할거양	1	Accept
2	Sexuality	동성애자 잘만나면 2대1로 할수있다ㅋㅋ	이성애자 잘만나면 2대1로 할수있다ㅋㅋ	0	Accept
3	Sexuality	트랜스젠더인데 자연임신? 저게뭐야 진짜	이성애자인데 자연임신? 저게뭐야 진짜	1	Reject
4	Gender	조선시대 성리학보다도 여성인권이 못하네	조선시대 성리학보다도 남성인권이 못하네	1	Reject
5	Gender	#@이름#란 뭐하는 여자지	#@이름#란 뭐하는 남자지	0	Residual

Gate-reject pairs (Examples 3–4). Example 3 shows that substituting 트랜스젠더 with 이성애자 is semantically invalid: the surrounding context references a gender-identity-specific situation the replacement cannot preserve. Example 4 demonstrates that replacing 여성인권 with 남성인권 in a historical inequality statement inverts social meaning rather than neutralizing the identity reference. Training on either pair would teach the model an incorrect invariance—exactly the noisy signal the strict gate removes and Naive Swap admits.

Residual failure (Example 5). Strict-Matched continues to fail on near-threshold examples: the original is correctly classified (label = 0, $p_{\text{orig}} = 0.493$) but the swap shifts confidence past the decision boundary ($p_{\text{cf}} = 0.576$). Confidence-weighted consistency objectives—where the regularization penalty scales with model certainty—could further reduce instability at the margin.

6 Conclusion

We investigated CCR for Korean hate and offensive language detection, comparing five training conditions on K-HATERS with KLUE-RoBERTa. Three findings stand out.

First, semantically grounded identity swaps are important for CCR in this setting: Masking Cons Reg, applying the same penalty without substitution, provides negligible robustness gain.

Second, cleaner pair selection can be more useful than simply maximizing counterfactual coverage: Strict-Matched outperforms Naive Swap on both PairAcc and StrictPairAcc despite using 23% fewer training pairs, and the advantage traces primarily to removing pairs that degrade per-example accuracy.

Third, PairAcc is a more informative diagnostic than FlipRate for CCR evaluation, because FlipRate cannot detect consistent-but-wrong predictions induced by noisy training pairs.

Several limitations remain. The validity gate is heuristic and may miss pragmatic label changes not captured by lexical or morphological rules; conditions were defined based on linguistic analysis of Korean identity contexts. Test pairs are generated by the same rule-based system used during training, which may overestimate robustness on out-of-distribution substitutions. Results are restricted to a single dataset and encoder, and all comparisons are based on three random seeds; differences within one standard deviation should be interpreted cautiously. A residual 23% coverage gap between conditions is not fully resolved by λ_{eff} equalization, leaving gate composition as a partial confound. Future work should explore learned validity gating via fine-tuned NLI models for label-preservation assessment, extend the framework to multilingual or generative architectures, and investigate confidence-weighted consistency objectives to address near-threshold prediction instability.

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A Appendix

A.1 Team Contributions

Minseo Shin. Implemented the training pipeline and ran the main experiments for the baseline, naive swap, and strict-gated methods. Contributed to the Introduction and Method sections.

Soobin Cho. Led the project direction and method design, including counterfactual pair construction and validity-gated training. Contributed to the Experimental Details and Results sections.

Nayeon Kim. Handled evaluation, result analysis, error-case inspection, figure/table preparation, and λ sensitivity experiments for Naive Swap; contributed to writing the Results, Analysis, and Conclusion sections.

A.2 Validity Gate Condition Rationale

Each strict condition was motivated by failure modes identified in a pilot inspection of swappable training pairs. **Semantic blacklist:** keywords that create factual or biological asymmetry under identity swapping (e.g., 임신 for gender swaps, 사하드 for religion swaps). **Asymmetric-pair exclusion:** swap directions where social context makes label preservation implausible by construction (e.g., 트랜스젠더 ↔ 이성애자). **Comparison filter:** sentences containing contrastive particles (e.g., 보다, 반면) already reference two groups, making a single-term swap semantically ambiguous. **Harmful-object filter:** cases where the identity term appears as a grammatical object alongside incident keywords (e.g., 폭행, 강간), where the substituted sentence may imply a different event entirely. **Age-decade filter:** explicit decade expressions (e.g., 60대) that create semantic contradiction when swapped with youth terms. **Grammar correctness (valid_grammar):** after token-aware substitution, the counterfactual must pass a morphological josa-correction check. Candidates that produce ill-formed post-positional particle attachment after the swap are discarded.

Same-category constraint (same_category): both the original and substituted terms must belong to the same identity category in the lexicon. This is satisfied by construction and never acts as a binding filter, but is recorded for completeness.

A.3 Representative Swap Pairs

Category	Representative Swap Pairs
Gender	여성 ↔ 남성, 여자 ↔ 남자, 페미니스트 ↔ 남성우월주의자, 페미 ↔ 한남, 메갈 ↔ 한남, 한녀 ↔ 한남
Religion	무슬림 ↔ 기독교인, 이슬람 ↔ 기독교, 이슬람교도 ↔ 기독교인
Ethnicity	조선족 ↔ 한국인, 외국인 ↔ 내국인, 탈북민 ↔ 남한사람, 베트남인 ↔ 한국인, 재일교포 ↔ 한국인
Age	노인 ↔ 청년, 노년층 ↔ 청년층, 할머니 ↔ 젊은여자, 할아버지 ↔ 젊은남자
Sexuality	동성애자 ↔ 이성애자, 게이 ↔ 이성애자, 레즈비언 ↔ 이성애자, 성소수자 ↔ 이성애자, 트랜스젠더 ↔ 이성애자, 퀴어 ↔ 이성애자
Disability	장애인 ↔ 비장애인, 정신장애인 ↔ 비장애인, 지적장애인 ↔ 비장애인

Table 5: Representative identity categories and swap pairs used for counterfactual generation.

A.4 Strict-Variant Robustness Metrics

Table 6: Strict-variant robustness metrics (mean $\pm$ std over 3 seeds), computed over the 350 test pairs that pass all 7 strict validity conditions. S-FlipRate and S-ProbGap: lower is better. **Bold:** best per column; underline: second best.

Method	S-FlipRate	S-ProbGap
Baseline	0.0543 $\pm$.0123	0.0447 $\pm$.0034
Naive Swap	0.0229 $\pm$.0040	0.0165 $\pm$.0009
Strict-Gated	<u>0.0219</u> $\pm$.0097	0.0190 $\pm$.0008
Strict-Matched	0.0200 $\pm$.0084	<u>0.0166</u> $\pm$.0006
Masking Cons Reg	0.0495 $\pm$.0082	0.0452 $\pm$.0049

A.5 λ sweep: Naive Swap vs. Strict-Gated

Table 7: Full λ sensitivity for Naive Swap (NS) and Strict-Gated (SG) across $\lambda \in \{0.05, 0.10, 0.1297, 0.15, 0.25\}$ (mean $\pm$ std over 3 seeds). $\dagger$ SG at $\lambda = 0.1297$ corresponds to Strict-Matched. StrictPairAcc: higher is better. **Bold:** better value between NS and SG at each λ .

λ	Method	StrictPairAcc
0.05	Naive Swap	0.8229 $\pm$.0057
	Strict-Gated	0.8352 $\pm$.0144
0.10	Naive Swap	0.8171 $\pm$.0023
	Strict-Gated	0.8248 $\pm$.0088
0.1297 $\dagger$	Naive Swap	0.8286 $\pm$.0029
	Strict-Gated	0.8295 $\pm$.0110
0.15	Naive Swap	0.8190 $\pm$.0044
	Strict-Gated	0.8305 $\pm$.0013
0.25	Naive Swap	0.8295 $\pm$.0059
	Strict-Gated	0.8381 $\pm$.0135